

Quiz day 6

1. Calculate the following complex values:

$$(6 - 3i) \cdot (2 - 5i), \quad \frac{1 - 8i}{3 - 4i}, \quad (1 - 3i)^3.$$

2. Consider the recurrence equation

$$a_n - 4a_{n-1} + 5a_{n-2} = 4n - 14.$$

Find the solution of the form $a_n = \alpha n + \beta$.